Remote sensing image object detection based on improved SSD

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***Abstract*—Aiming at the problem that the detection accuracy of the SSD object detection model is insufficient when detecting remote sensing images, an improved SSD object detection model is proposed.This method uses Mobilenet v3 to replace the backbone network of the ssd algorithm, and adds the FPN network for feature fusion, and the shallow feature map with high resolution is fused with the deep feature map with rich semantic informationAt the same time, a feature pyramid is built between feature maps to enhance the target feature. Experimental results show that the detection accuracy of the improved method is significantly improved compared with the original algorithm, and the effect on the detection of remote sensing image objects is better.**

Keywords: Remote sensing images; Deep learning; SSD; Object detection; FPN

I. INTRODUCTION

With the rapid development of China's aerospace technology, the continuous progress of remote sensing technology, remote sensing image acquisition is more convenient and efficient than before, remote sensing image has the characteristics of large data volume, rich variety, clear imaging, high timeliness, in line with the development trend of today's social big data era, therefore, in the military and civilian fields, remote sensing image object detection has been widely used.

The initial convolutional neural networks didn't work well and were difficult to train, and in 2012, Alexnet[1] proposed by hinton's team in the Imagenet image recognition competition introduced a new deep structure and dropout method, reducing the error rate from more than 25% to 15%.Since then, a series of new network structures such as VGG[2], ResNet[3], GoogLeNet [4-5], DenseNet[6] and so on have emerged, all of which have achieved excellent classification and recognition effects.At present, the object detection algorithm based on deep learning can be divided into two categories from the overall architecture, namely two-stage detection algorithm and single-stage detection algorithm.The two-stage detection algorithm divides the object detection process into two parts: generating a candidate calibration box for a given target and identifying the categories of targets within the frame.Based on the existing research, the false detection rate and missed detection rate of the two-stage detection algorithm are very low, and the detection accuracy rate can reach a very high level, while the disadvantage is that it needs to be repeatedly detected and classified, and the detection speed is slow.The classical algorithms of the two-stage detection model include R-CNN, Fast R-CNN, Faster R-CNN[7], Mask R-CNN[8] and so on.The single-stage detection algorithm summarizes all the processes into one branch, and the detection results can be obtained directly, so it is faster than the detection speed of the two-stage detection algorithm; the classic algorithms of the single-stage detection model include the YOLO[9] series and the SSD[10].

The development of natural image object detection has also promoted the optimization of remote sensing image object detection, but remote sensing images are different from natural scene image imaging methods, and remote sensing images have their own unique characteristics, such as: high resolution, small and dense targets, complex backgrounds and so on.Therefore, further targeted improvements to the migration method are needed to solve the difficult problems in remote sensing image object detection. In this paper, the SSD object detection model is improved by combining Mobilenet v3 and FPN[11], and the phased IOU threshold is introduced, which improves the accuracy of target detection of remote sensing images by strengthening the extraction of target features and fusing the position information of shallow feature maps and semantic information of deep feature maps.

II. TRADITIONAL SSD OBJECT DETECTION MODEL

SSD object detection model is a very good One-Stage object detection model, SSD mainly through the convolutional neural network for feature extraction, and then uniformly sampled in different positions of the picture, sampling will be preset different scales of the aspect ratio, object classification and prediction box regression at the same time, the whole process only needs one step, its main advantage is fast.

The SSD model takes VGG16 as the backbone network, and modifies it to a certain extent, converts the FC6 layer and FC7 layer of VGG16 into a convolutional layer, replaces it with two convolutional layers with convolutional kernels of 3*3 and 1*1, and removes all Dropout layers and FC8 layers, adding four convolutional layers of Conv6, Conv7, Conv8, and

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Conv9, realizing the full convolution of the network. The backbone network is shown in Figure 1.

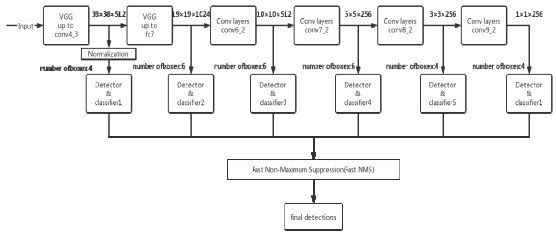


Figure 1. SSD object detection model

As can be seen from Figure 1, the SSD detection model takes the features of the third convolution of conv4, the characteristics of the fc7 convolution, the characteristics of the second convolution of conv6, the characteristics of the second convolution of conv7, the characteristics of the second convolution of conv8, and the features of the second convolution of conv9 for the next step.

The number of prior boxes corresponding to each feature point of each feature layer is 4, 6, 6, 6, 4, 4, respectively. For each feature layer obtained, it is also necessary to perform two convolution operations to predict the change of each prior box at each mesh point on the feature layer and the corresponding kind of prediction on each mesh point on the feature layer.

After convolution, the resulting x_offset, y_offst, h, and w are adjusted, where the x_offset and y_offst represent the offset state of the real box from the center of the a priori box, and h and w represent the change in the height and width of the real box relative to the prior box. Then the center point of each grid is added with the corresponding x_offset and y_offset, the result is the center of the prediction box, and the a priori box is adjusted by h and w to obtain the width and height of the prediction box, and finally the score sorting and non-maximum suppression filter of each prediction box is performed to determine the category to which the objects in the prediction box belong.

III. IMPROVEMENT BASED ON SSD MODEL

A. *Construction of the backbone network*

The original SSD network adopts VGG16 as the backbone network, the overall structure is relatively simple, and the transmission direction is single, which cannot fuse the information between different layers, and there are some shortcomings in feature extraction. To this end, the improved method proposed in this paper adopts Mobilenet-v3 as the backbone network to replace the backbone network VGG16 of the original algorithm.Mobilenet-v3 is a lightweight attention model that combines the deep separable convolution of MobileNet-v1 with the inverse residual structure of MobileNet-v2 with a linear bottleneck, using h-swish as the activation function.The h-swish activation function formula is as follows:

$$\mathrm{h}\text{-}swish(x) = x \cdot \frac{RELU6(x+3)}{6} \tag{1}$$

In the process of feature extraction, several bneck structures are passed, which are shown in the following figure.

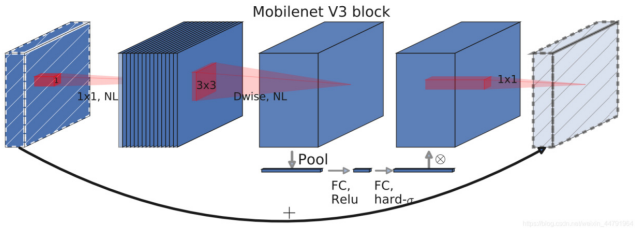


Figure 2. Bneck structure

The bneck structure first uses 1*1 convolution to ascend dimension, and then performs 3*3 depth separable convolution, adjusts the weight of each channel through the lightweight attention mechanism, and uses the h-swish activation function instead of the swish function to reduce the amount of operation and improve performance.

B. *Phased IOU threshold*

The SSD detects the target through the prior box, and in the process of training, the type of the prior box belongs to is first determined by the IOU of the prior box and the real box.Typically, the a priori box threshold for an SSD model is set to 0.5. But the way in which the iOU threshold is fixed to filter the a priori box can have an impact on targets of different scales.In SSD object detection, a total of 6 feature maps are extracted from conv4_3, fc7, conv6_2, conv7_2, conv8_2, conv9_2, and a priori box of different scales is set on each feature map, and the calculation method of the scale of the prior frame scale is shown in Formula (2).

$$\mathrm{s}_k = s_{\min} + \frac{s_{\max} - s_{\min}}{m-1}(k-1), k \in [1, m] \tag{2}$$

From the above formula, it can be seen that with the size of the deep feature map and its transcendental frame is also increasing, the targets that need to be detected in the remote sensing image usually account for a small proportion of the entire map, and the detection target will be smaller than the size of the prior frame, compared to the target of the larger scale, the small target is more difficult to meet the condition of IOU greater than 0.5, resulting in a small number of selected small targets and insufficient training. This paper solves this problem by setting different IOU values for prior boxes of different scales

C. *FPN*

Multi-scale target recognition is of great significance in object detection and classification. Traditional detection and recognition methods typically solve this type of problem by building image pyramids. In deep learning, FPN networks are modeled on the structure of image pyramids, which is significant for multi-scale object detection. Among them, the information fusion of FPN to the shallow feature map of the model can improve the robustness of small target detection.

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Figure 3. FPN structure

The implementation of the FPN network is divided into two steps: first, the feature pyramid is constructed with the feature map of different convolutional layers; second, the feature map of the last layer is upsampled and fused with the previous layer feature map to generate a new feature pyramid. The network can change the connection of the convolutional layer in the network model while maintaining the original computation amount, and integrate the features of deep semantic richness into the shallow layer, so that the shallow semantics are more abundant, and the detection accuracy of small-scale targets is greatly improved.

FPN networks employ a bottom-up, then top-down structure. Bottom-up refers to the use of convolutional operations to generate feature maps to build feature pyramids, and from top to bottom, it is from the top feature map of the feature pyramid to sample sequentially and then fuse with the shallow feature map to obtain a new feature pyramid, which makes the FPN have a great improvement in the effect of the detection network.

IV. EXPERIMENTS AND ANALYSIS OF RESULTS

A. *Experimental platform and dataset*

The experimental part uses the RSOD-Dataset dataset, which was annotated by the Team of Wuhan University and has a total of 976 optical remote sensing images, including four types of targets: aircraft, playground, overpass, and oil drum. In order to avoid overfitting during training, the data is enriched by transforms such as rotation and flipping. During training, 70% of the dataset is used for training and 30% for testing. The experimental results were compared with the original SSD algorithm to verify the effectiveness of the improvement.

B. *Experimental evaluation criteria*

In this experiment, mAP (Mean average precision) is used to evaluate the performance of improved SSD algorithms.mAP is a commonly used metric in object detection algorithms to measure the accuracy of algorithms, involving precision and recall.Precision refers to the proportion of related categories in the returned results to the total returned results. Recall refers to the proportion of the relevant categories in the returned results to the total returned results. Accuracy and recall are defined as follows:

$$\text{precision} = \frac{TP}{(TP+FP)} \tag{3}$$

$$\text{recall} = \frac{TP}{(TP+FN)} \tag{4}$$

where TP is the number of related categories in the returned results, FP is the number of unrelated categories in the returned results, and FN is the number of related categories that are not returned.

From the correctness rate and recall rate, the AP curve of each category can be found, and then the average of the APs of all classes can be averaged to obtain mAP:

$$mAP = \frac{\sum_{k=1}^{N} P(k)\Delta r(k)}{m} \tag{5}$$

where N represents the number of images in the test set, and P(k) represents the accuracy value of the k-piece image recognition,r(k) represents the amount of change in the Recall value when the number of recognition images varies from k-1 to k,m is the number of categories for all images.

C. *Experimental results*

In order to verify the effectiveness of the improved SSD object detection model in detecting targets on optical remote sensing images, the original SSD object detection method and the improved method are trained and tested on the RSOD-Dataset dataset, and the detection results are shown in Table 3:

TABLE I. THE DETECTION ACCURACY OF DIFFERENT OBJECTS BY DIFFERENT METHODS

	plane/ %	oil tank /%	playground /%	overpass/ %	mAP/ %
SSD	89.07	90.90	72.95	51.16	76.02
This article algorithm	92.27	92.49	82.74	54.62	80.53

As can be seen from Table 3, the improved SSD target detection model mAP is 4.51% higher than the original SSD model. Figure 4 is part of the renderings of the remote sensing image detection results, the rectangular box in the figure is the position of the target predicted by the model, figure (a), figure (c) is the detection result of the picture after the SSD target detection model, it can be seen that the detected aircraft and oil drums have low scores, and some of the objects with small targets in Figure (a) cannot be detected, Figure (b) and Figure (d) are the detection results of the picture after our improved SSD model, and some of the targets that have not been detected by the original SSD target detection model can be detected , and the score is higher.



(a)SSD (b) Improved SSD

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(c)SSD (d) Improved SSD

Figure 4. Remote sensing image object detection rendering

Usually the size of the prior box in the SSD model is 30,60,111,162,213,264,315, due to the small target in the remote sensing image, this experiment sets the prior box to 21,45,99,153,207,261,315, and at the same time sets the staged IOU threshold to make the training more adequate, and the phenomenon of repeated testing appears in part of the target in figure (a) in the figure below, and the improved network training is tested after the test, Each object has only one box to label it and the score is improved. Combined with Figure 4 and Figure 5, it can be seen that the improved network can better extract the characteristics of the target, making the target detection effect better.



(a) SSD (b) Improved SSD

Figure 5. Remote sensing image object detection rendering

V. Conclusions

In order to solve the problem that the detection accuracy of the SSD object detection model on the remote sensing image is insufficient, this paper proposes an improved SSD object detection method.First of all, Mobilenet-v3 is used as the backbone network to replace VGG16, which has a deeply separable convolutional and inverse residual structure, which has a better effect on the feature extraction of the target during training.The phased IOU threshold is adopted, which can make the model train the small target more fully, and at the same time, the FPN network is added, so that the features of different convolutional layers in the network model are fused, and the features rich in deep semantics are integrated into the shallow layer, so that the accuracy of target detection is significantly improved.Experimental results show that the proposed method can effectively detect the target in the remote sensing image, reduce the missed detection rate of the target, and realize the efficient detection of the target in the remote sensing image under complex background.

References

[1] KRIZHEVSKY A,SUTSKEVER I,HINTON G E. Image Net classification with deep convolutional neural networks[C]. Advances in Neural Information Processing Systems. Harrahs and Harveys, Lake Tahoe: NIPS Foundation, 2012:1106-1114.

[2] SIMONYAN K,ZISSERMAN A.Very deep convolutional networks for large-scale image recogn- ition[C]. International Conference on Learning Representations.San Diego, CA, USA: FAIR, 2015.

[3] HE K, ZHANG X, REN S, et al. Deep residual learning for image recognition[C]. Proceedings of IEEE International Conference on Computer Vision and Pattern Recognition. Las Vegas, USA: IEEE, 2016:770-778.

[4] SZEGEDY C, LIU W, JIA Y Q, et al. Going deeper with convolutions[C]. Proceedings of IEEE International Conference on Computer Vision and Pattern Recognition.Boston,USA: IEEE,2015:1-9.

[5] SZEGEDY C, VANHOUCKE V, IOFFE S, et al. Rethinking the inception architecture for computer vision[C]. Proceedings of IEEE International Conference on Computer Vision and Pattern Recognition. Las Vegas, USA: IEEE, 2016:2818-2826.

[6] HUANG G, LIU Z, MAATEN L, et al. Densely connected convolutional networks[J]. ar Xiv preprint ar Xiv:1608.06993, 2018.

[7] REN S Q,HE K M, GIRSHICK R,et al.Faster R-CNN:towards real-time object detection with region proposal networks[J].IEEE Transactions on Pattern Analysis and Machine Intelligence,2017,39(6):1137-1149.

[8] HE K, GKIOXARI G, DOLLA P, et al. Mask R-CNN[C]. Proceedings of IEEE International Conference on Computer Vision. Venice, Italy: IEEE, 2017:2061-2969.

[9] REDMON J,DIVVALA S,GIRSHICK R,et al.You only look once:unified,real time object detection[C]Proceedings of IEEE Conference on Computer Vision and Pattem Recognition.Washington D.C.,USA:IEEE Press,2016:779-788.

[10] LIU W, ANGUELOV D, Dumitru Erhan. SSD: single shot multi-box Detector[C]. Proceedings of European Conference on Computer Vision. Amsterdam, Netherlands: Springer, 2016: 21-37.

[11] KIN T, DOLLAR P, GIRSHICK R, et al. Feature pyramid networks for object detection[C]. Proceedings of IEEE International Conference on Computer Vision and Pattern Recognition. Hawaii, USA: IEEE, 2017:2117-2125.

[12] Tian Z,Shen C, Chen H,et al.FCOS: Fully Convolutional One-Stage Object Detection[J]. Computer Vision and Pattern Recognition,2019.

[13] Lin T,Dollar P,Girshick R,et al.Feature Pyramid Networks for Object Detectors[C].Computer Vision and Pattern Recognition,2017:936-944.

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